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CHUNG-CHING HUANG, Indiana University Bloomington, Bloomington, IN, United States

ERIK STOLTERMAN, Indiana University Bloomington, Bloomington, IN, United States

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Capturing Prolonged Interactive Experiences: A Review of Visual Approaches in User Research

Chung-Ching Huang
Indiana University
919 E 10th Street
Bloomington, IN 47401
huang21@indiana.edu

Erik Stolterman
Indiana University
919 E 10th Street
Bloomington, IN 47401
estolter@indiana.edu

ABSTRACT
Each user of interactive systems experiences prolonged engagement with applications, products and systems. Because of limitations of contemporary long-term user research methods and imperfection in human memory, approaches for longitudinal user experience are still under-explored. In this paper, we review a recent methodological shift in user research aimed at applying visual approaches to the capturing of prolonged interactive use. We analyze and discuss selected examples of existing visual approaches that support the examination of longitudinal use and represent and visualize changes in interactions. We discuss potential merits of visual approaches for user experience longitudinal research. We also discuss the drawbacks of these approaches. Overall, we argue that visual approaches have the potential to support certain aspects of user experience research, especially in relation to issues concerning temporality.

Author Keywords
Use experience; visual approaches; user research; temporality.

ACM Classification Keywords
H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

General Terms
Design; Theory.

INTRODUCTION
Longitudinal user research has been widely applied in the field of HCI aiming at capturing long-term interactive use within products or systems. Still, HCI researchers have encountered difficulties in long-term user studies, such as choosing proper temporal scales in experiments, analyzing in-situ use data, or making sense of the information derived from users' memory. Although these are ongoing research challenges for longitudinal user studies, we believe that long-term use experience and users' retrospective insights do consist of valuable information. In the mean time, prior theoretical initiatives such as describing an interactive experience as a trajectory [4] or as an adapting process [18] also draw our attention to temporal issues and prolonged interactive use. As a result, we truly believe that *use experience over time* is a potential rich source for future HCI practice and research.

Based on the predisposition mentioned above, our research project starts with how we reconsider suitable procedures to improve longitudinal research, and how those procedures facilitate understanding use experience situated in the past, through now, and to the future (see figure1). This agenda might cover a wide variety of critical research problems. Instead of investigating each potential aspect of temporal issues in HCI, this paper focus especially on prior studies that explore the following two research questions: first, how *users* might recall their prior interactive use, and second, how *researchers* can tackle data derived from longitudinal research.

In addition, we have observed an emerging trend of introducing visual representations and data visualizations aiming toward the two issues mentioned above. Data visualization has benefits when it comes to presenting temporal information, such as locating use data and presenting them along within the use timeline. However, when we take a closer look at longitudinal research practices, we find that visual approaches serve purposes beyond merely representing data. We have observed several innovative visual approaches, which we will further introduce in this paper, that enable *users* to *retrospect and reflect* on their prior interactive use while also enabling *researchers* to *analyze* less structured and fragmented use information. Our preliminary insight was echoed when we participated in workshops exploring new methods and tools for temporal issues in HCI and interaction design [15,25], where we learned that many colleagues have started applying visual approaches and visual thinking in their users studies.

In short, our research is aimed at *examining if visual approaches might help users and researchers to explore and study critical temporal issues in longitudinal research*. To further explore the use of visual approaches, we first survey current difficulties in longitudinal user studies. We also highlight challenges and potential benefits of research

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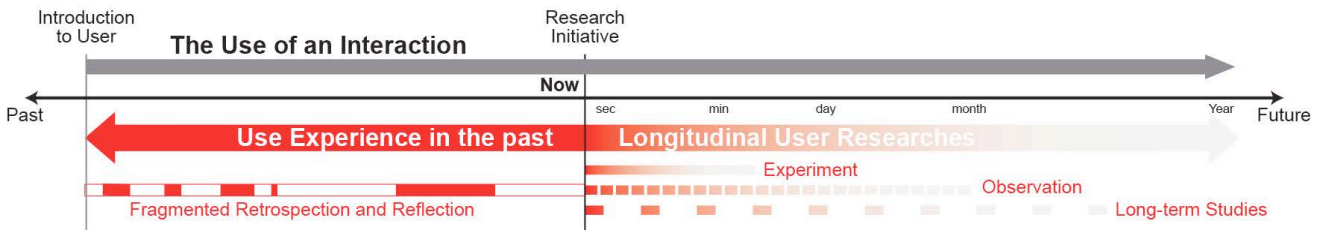


Figure 1: Perspectives we concern in longitudinal research methods

that focus on retrospection and reflection. Through the careful selection of innovative research examples, we examine if and how visual approaches can help to overcome some of the current longitudinal obstacles. Finally, we discuss possible benefits and drawbacks of visual approaches. We end by arguing that such approaches might offer suitable supplemental perspectives in future longitudinal user research.

STUDY PROLONGED INTERACTIVE USE EXPERIENCES

Through a literature review of the advantage and shortcoming of current longitudinal methods, we first discuss some difficulties with current research and, in the later sections, how visual approaches might be an improvement.

Researchers in HCI have been introducing various approaches and procedures to understand prolonged interactive use experience. These research projects have focused on different time scales, from seconds and minutes to months and even years. Researchers have also proposed various theories toward conceptualizing long term use experience [18]. Even with those fruitful research results, HCI and interaction design researchers still encounter difficulties. Some challenges include how to define a proper temporal scale in research or how to select suitable methodologies to study prolong interactive use. We will in the following paragraphs briefly describe some re-occurring issues within longitudinal research.

First, each research approach has its unique strength and potential weakness regarding long-term use experience. For example, when researchers investigate interactive use they have to recruit users to complete certain defined tasks in a lab experiment setting. Such tasks are well structured and can be finished in a relatively reasonable period of time, such as in seconds or in minutes. The advantages of those prescribed research experiments is that, with contemporary research procedures such as video recording, direct observation or logging software, the interactive use can be closely scrutinized. The data captured can be analyzed in a continuous and comprehensive manner. However, studying an interaction task that might last for hours, days or months, might generate enormous amounts of data that would require a lot of effort to analyze, which is often beyond the scope of many research projects. Also, direct observations in the research setting often influence the subjects' actual use which impacts the findings of the research [1,8]. As a result, continuous logging, observation and detailed analysis with prescribed experiment design might not be applicable for all long-term interactive use scenarios.

Second, interactive use happen not only in the research lab environment but most of the time in everyday use context. As a result, many research methods have aimed at studying in-situ prolonged use [12], and some examples of those research methods are experience sampling method (ESM) [9], day reconstruction methods [14], diary study [24], culture probe [7], or a combination of several procedures like data logging plus online diary [21]. With these methods, users are supposed to report and reflect on their use experiences with oral or written text descriptions several times a day. These methods allow researchers, without intense software logging or direct observation, to get an overall understanding of individual prolonged interaction. However, the gathered information is inherently fragmented because collected use data are naturally discontinued in time and space, compared to data collected in a lab environment.

Last but not least, it is less applicable to use the same research approaches, such as log analysis or direct observation, to investigate use scenarios that *already occurred in the past*. The reason is quite straightforward: we cannot go back to the previous use case in prior space and time, to research interactive use experiences with approaches originally designed to understand what is happening right now. For example, experiments can only be conducted after the researchers raised specific research questions and has set up a relevant experimental context. Also, many prior interaction use cases are unique and not always reproducible in the current use context. In order to grasp a glimpse of what happened in the past, researchers rely on user's own retrospection in interviews, focus groups or survey studies. Unfortunately, since people have imperfect memory to recall correctly every detail of their prior interactive use, the use information collected is fragmented, sometimes even without accurate chronicle order, and might be considered less convincing compared to data collected with well defined experimental studies.

To conclude, we find that current research approaches have successfully captured various long-term, in-situ use experiences. More importantly, many significant interactive scenarios and use experiences happen in the past, and we heavily rely on user's own retrospection and reflection with

their imperfect memory to attempt to understand these past interactions.

ISSUES OF RETROPECTION AND REFLECTION WITHIN LONGITUDINAL USER RESARCH

In this section, we further discuss how users recall and reflect in interviews, questionnaires, and focus group settings. The format of research records and how researchers ask questions might impact how they can perform recall and retrospect well of their longitudinal use.

Recall activities in current user researches

In many user studies, participants are requested to comment on their prior interactive use experiences [6][22]. Aided by structured or open questions, participants recall prior interaction scenarios and reflect on those use cases. However, researchers design research questionnaires based on their assumptions and the goals of the research, such as revealing various product values including utility, usability, beauty, identity, stimulation, with a structured interview technique CORPUS [26]. Researchers pre-define options and scales in the questionnaires or structured interview; as a result, subjects are not able to provide further information beyond what the researchers have already framed as problems and anticipated answers.

In many cases, users report only their initial impressions in an interview or questionnaire, for example, the enjoyment of using this interaction [8]. In order to assist users to recall a specified moment in the past, extra aids might be needed in the interview or survey. For example, day reconstruction method (DRM) is a research procedure providing chronological aid for retrospection within one day, which allow the subjects to focus on a period of 24 hours [14]. However, even with the help from available research techniques users still have only a vague concept about their trajectory of use in the past. This is especially true when comparing data gathered from tasks in a predefined use experiment.

Focus groups are another survey method, by which researchers recruit multiple people and ask the participants to discuss their experiences in a group setting. Participants can discuss their previous interactive use cases among each other. Research facilitators can ask questions or request clarification from participants' conversation to help the participants to recall their interactions [20]. A good prior use case often triggers an active discussion in focus group research; however, not all the subjects recall and express their opinions in a similar fashion, which may stimulate vigorous discussion or may impinge on discussion. Further, not all the participants talk about their experience in a correct temporal order. Their thoughts might leap forward and backward in time and space. As a result, the data gathered in a focus group research are often discontinuous in space and context, which makes systematical analysis of interactive experience, especially from a temporal perspective, even more difficult.

Format of retrospection and reflection

People reflect on their own previous use cases and describe their prior experience in unique ways. For example, ordinary end users might write or talk about their use experience with expressions such as "like" or "dislike" without any detailed reflection. Design critics, experience designers or design researchers might articulate their own use experience with theories or advanced insights, such as when Buxton compares the juicers he has used and his experiences involved [5]. Similar issues happen when an individual research participant describes his use experience with his own words in the interview and focus group study. Researchers face increased difficulty in analyzing and comparing data drawn from these methods because of the varying degrees of description, in breadth and in depth, of different participants.

Retrospective use information collected by survey or interview research comes in various forms and styles. Researchers need to compare different descriptions from users in some systematic way. In order to help the participant recall their prior use with better chronicle framing or order, some supplemental methods might be needed.

CRITERIA OF CHOOSING EXAMPLES OF VISUAL APPROACH IN THIS RESARCH

Based on some criteria we carefully selected some visual approaches for our review. These criteria can be seen as responding to the insights gathered from our literature review and the analysis of longitudinal research methods. We were selecting visual approaches that might:

1. help the research participants to better *define temporal framing and chronicle orders*,
2. help the user to *describe particular prior user experiences*, which are not yet easily achieved by text or oral descriptions,
3. enable better *discussion and communication* during the research process,
4. help researchers to *analyze and/or present research data and findings*.

We did not decide which approaches to include based on whether these visual approaches are widely applied, or whether user studies with these visual approaches have been reported and recognized. Neither do we intend to argue that the chosen visual approaches are the best solutions for all future longitudinal users study.

Instead, we have engaged in a critical assessment of if and how these procedures can advance our understanding of prolonged use experience in human computer interaction.

EXAMPLES VISUAL APPROACHES IN LONGITUDIANL USER STUDIES

In the following section, we briefly present some existing approaches. We also present some of our own earlier work that has been published elsewhere.

We have arranged the examples into two categories according to how they serve different roles in user studies: *users* and *researchers*. The first group of visual approaches is focused on how users can provide retrospection and reflection input. The second group of visual approaches is aimed for researchers to support their analysis and representation of data regarding longitudinal and retrospective use.

We do not evaluate the approaches; instead our approach here is analytical in the sense that we use the examples to review what temporal aspects have already been tried and to get insights on what is potentially possible.

Visual approaches designed for users

Here, we choose four visual approaches that all include diagrams or two-dimensional coordination. Each approach deals with different temporal framing, and all of them focus on particular aspects in the overall use experience.

iScale

With the research software iScale [17], the participants are requested to "sketch" the use experience from the moment of purchase until now (see Figure 2). With the interface, participants can record an event and their opinion, such as "more easy to use / more difficult to use" during the use after the purchase. Meanwhile, a text description is recorded regarding the value of the event such as how significantly this event influences the overall experience and how confident the participants feel about the detail. As a result, users mark prior use events in a proper *chronicle order*, and through connecting those events they will see the *overall temporal framing,* as well as the general trend of their own use in the past.

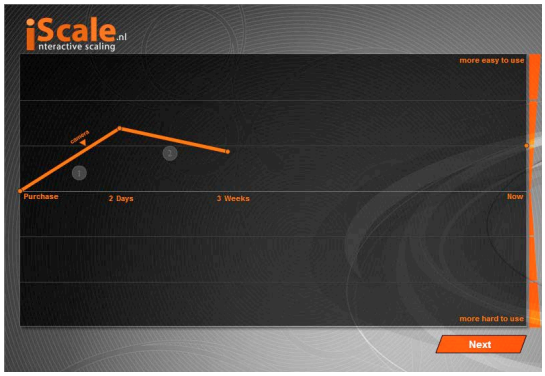


Figure 2: iScale [17]

UX curve

In the research of long-term mobile use experience, the UX curve [19] is introduced with a template, including a diagram representing specific measurement over time and a reserved area for participants to write down text description (see Figure 3). In the diagrams, a light grid is provided to hint the level of measurement of previous use experience, such as general relationship, the attractiveness, ease of use, utility, and the degree of usage within mobile devices. Participants are requests to draw a curve on the diagrams; further, they also annotate on the curve and explain the graph.

The tendency of use is easy to trace in each diagram. The events on the diagrams also follow a pretty *accurate chronicle order*; however, it is not represented in a precise temporal scale on the coordination. The coordinate of the event does not reflect the actual timing in the past.

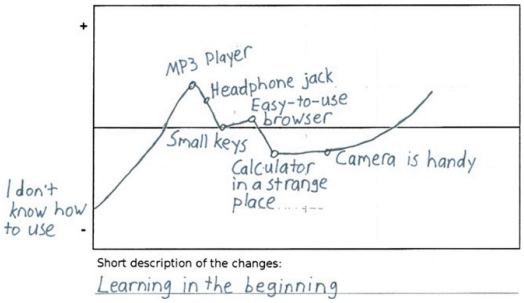


Figure 3: UX curve [19]

Retrospective reconstruction sketch

In the final stage of a three phases user research, the retrospective reconstruction sketch template (see Figure 4) are provided in an user interview [13]. After the first stage of initial use and second stage of three weeks diary study, research participants draw on this diagram and recall specifically three measurement of use experience: innovativeness, ease of use, and usefulness. Since users have been involved in prior three weeks interactive use, *the temporal scale is fixed on the diagram that is three weeks*. The prior use time is evenly distributed on the dimension. User's diary in phase two served as memory clues to sketch the curve of measurement over the dimension. Annotation is also encouraged on the diagram.

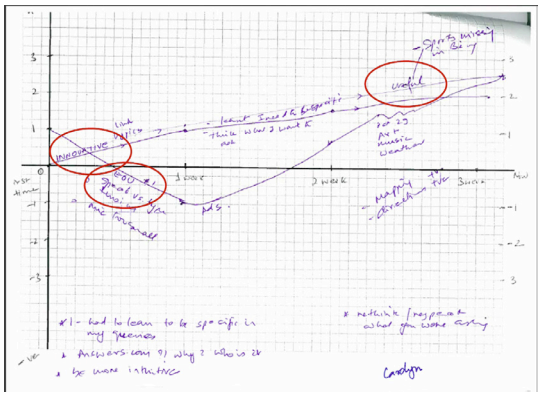


Figure 4: retrospective reconstruction sketch [13]

Sketch for a interactive use in a common day

In our early research [10] we focus on how users think about their common use of email. Before we conducted the interviews, we asked the participants to recall how they paid attention to email usage in any ordinary weekday. The participants had to think first about their previous use scenarios situated in twenty-four hours such as weekday or weekend, and draw the level of attention on the diagram (see Figure 5). After the sketching session, the users were asked to explain their sketch to researchers, special attention was given to explaining the meaning of trajectory changes within the sketch.

The variety of drawing styles to represent prior use scenarios were beyond our expectation, even with only a small number of research subjects in our preliminary study. Meanwhile, the research facilitator asked the participants, in the follow up interviews, to explain their sketch of email use on the diagram. The diagram became an aid for the participants to recall their patterns of use within one particular day. The diagram enables the research participants to reflect as to why their attention level increased or decreased throughout an ordinary day. The interview protocols are also analyzed as comparison.

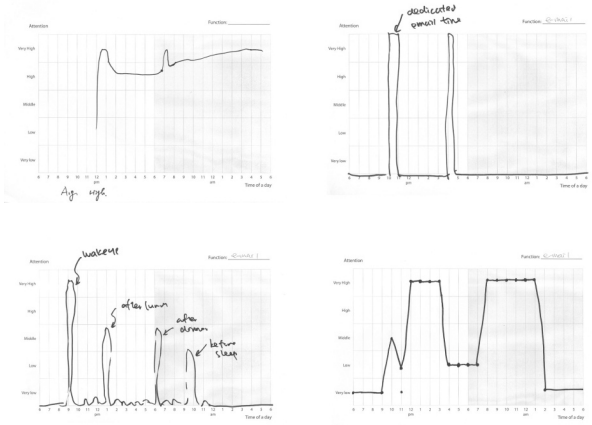


Figure 5: examples of drawing email usage attention level [10]

Discussion

In these four examples, we see a visual fashion of sketch, thought and reflection over previous use experience. Different diagrams are designed and applied toward various *measurements of use experience*. An aspect of measurement is purposely chosen for each diagram, for example, *easy or difficult to use* in [17], *general relationship, the attractiveness, ease of use, utility, and the degree of usage* in [19], *innovativeness, ease of use, and usefulness* in [13], and *attention level using email* in [10]. Users might encounter difficulties describing "measurement of use" directly with text or oral description; however, within a diagram they can mark an event or sketch a trend intuitively. All of the diagrams keep a chronicle order, but *the scales of time* in the diagrams are quite different, such as from purchase to now, exactly in the past three weeks, in a common weekday, or in the past.

Overall, the examples show that there are attempts to use visual form of retrospective user input as an alternative approach aimed at capturing prior use experience with *proper temporal scale and framing*. Those visual ways also help make users aware of *proper chronicle order*. Discrete events on the diagram can be discussed and annotated, and the linkage of events represents an emergent trend of previous use.

Visual approaches designed for researchers

In the following section, we present some examples of visual approaches designed for researchers. The examples included here have mainly two purposes: first, they are aimed at supporting researchers to *analyze* longitudinal data from users, and secondly to help researchers to *represent* trajectories of change of prior use.

Analyzing users' sketch of longitudinal use

Karapanos et al. have identified visual elements form users' sketches about longitudinal use, which are called *segments* [16]. Segments are categorized into three different types of *reports* (discrete experience, overall evaluation with reasoning through the experience, and overall evaluation with no further reasoning) and four different types of *sketches* (constant, linear, non-linear, and discontinuous). They have also developed micro use concepts in a visual fashion. These concepts help researchers *analyze* each sketch from users (see figure 6).

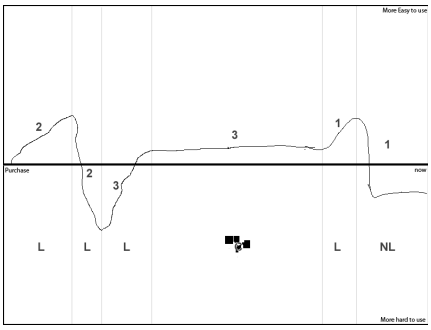


Figure 6: free hand sketch and identify the segment [16]

In our early research of attention level in email usage [10], we also define, using the similar approaches applied in Karapanos et al.'s research, the concepts of use and visual elements within user's sketches. We have first prosed a preliminary set of graphic annotations that can be used to analyze users' sketches (see Figure 7). For example, each rise of attention in the sketch can be described as a *session* of use, and every session has its own *duration* of use. There is *interval* between two sessions, and the total number of sessions in the period determines the *frequency* of use. Since we conducted open interviews with users based on their own sketches, we also propose coding schema to analyze users' interview transcription, such as *elements of*

time, *focus*, *attention*, *action*, *notification*, and *intention of use*.

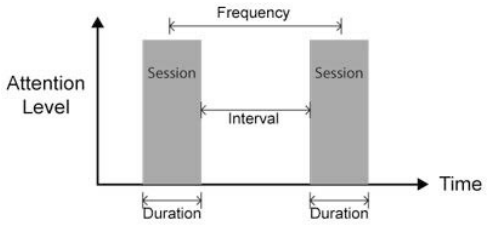


Figure 7: graphic annotation [10]

With both visual annotations of use and interview protocol analysis, we analyze those six email use cases and conclude with two emerging patterns: with a plan and with no plan. We also showcase the use patterns in a table, in which both visual representation and text description are included (see table 1). This table showcases how we combine visual annotations over time and analyzed interview protocol to represent longitudinal email use cases.

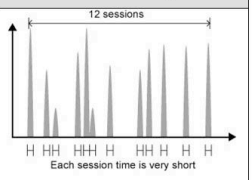
Pattern A: with plan	Pattern B: no plan
	
User **preplans** 2 **sessions** of email usage with **high focus** Unfinished work will be **postpone** to next session During the **interval**, he will turn off the **notice and warning** to **stay off-focus** If he finds some empty time beyond his daily **schedule**, he will engage in an **opportunistic** new session	User does not **plan** for email usage The email notice will **trigger** the **attention and a session.** He **judge** weather to **start** a **section** by • Review mail title • See current **schedule** • See current activities can be **interrupted** or not The **length** of each **session** is short

Table 1: two email use patterns [10]

Represent the history and the trajectory of interactive use

Regarding changes in the past within interactive systems, we choose two research examples, in which visual approaches serve more than just the display of temporal change. Researchers reflect critically over core questions or concepts, with selected visual elements, in their study toward prolonged interactive experience.

In service design research and practice activities, *customer experience journey maps* or *service blueprints* have been widely adopted. Such visualizations describe the critical events in a service; furthermore, the mapping emphasizes critical aspects of service, such as how users navigate across various sectors of service. Many of them are also described with implicit chronicle order, such as from the early adaption to the matured current use of the service.

Polaine et al. propose a new visual approach, BLUEPRINT+, to foreground the thinking activities using service blueprints [2]. First, BLUEPRINT+ nicely layouts all interactive activities, along with the timeline with *correct chronicle order* and *time indication*, regarding various actors in a service journey. Further, the solution that the authors proposed adds extra layers of information in the diagram, such as the emotional states in the bottom part of figure 8, to facilitate the planning of service.

The obvious advantage of this BLUEPRINT+ visual approach is that it represents multiple layers of information effectively in one diagram. In addition, the extra layers of information, which is the emotional state over time here, enable service developers to compare and reconsider emotional relationships among various actors involved in the service. The visual layout enables an effective reference of desired core concepts of this service design.

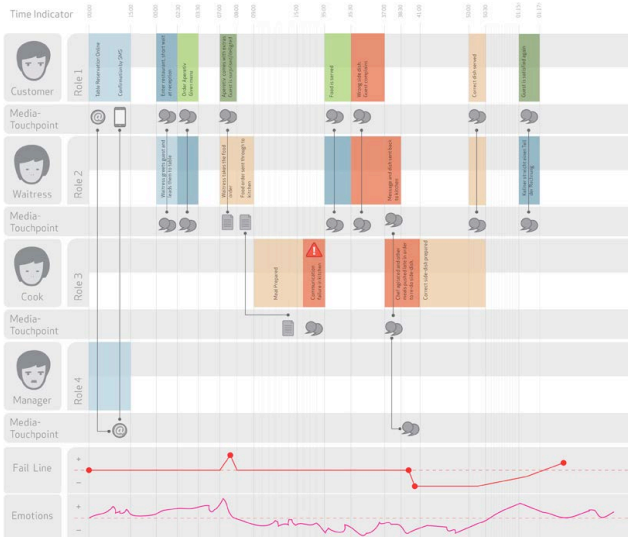


Figure 8: Emotions as an extra layer in BLUEPRINT+

In our prior research for the history of mobile device interactive use, we conducted analytic review for mobile photography in the mobile phone usages in a visual way [11]. We intended to explore notions of interaction critique through the construction and analysis of visual narratives in research design.

In our study, we first apply an artifact-centered perspective, as our core concept. We limited our focus to exploring the function, interface and a goal of the interactive product. We also present this concept in a form of visual schema (see figure 9): on the left are the internal function and external interface, and on the right are the goals of the interactive products. The change of function or interface is indicated through solid color, and the changes of goal are also marked with colored areas on the right. The green ones indicate a designed change, and the red one indicates user adaption. We analytically describe the trajectory of the changes of mobile photography.

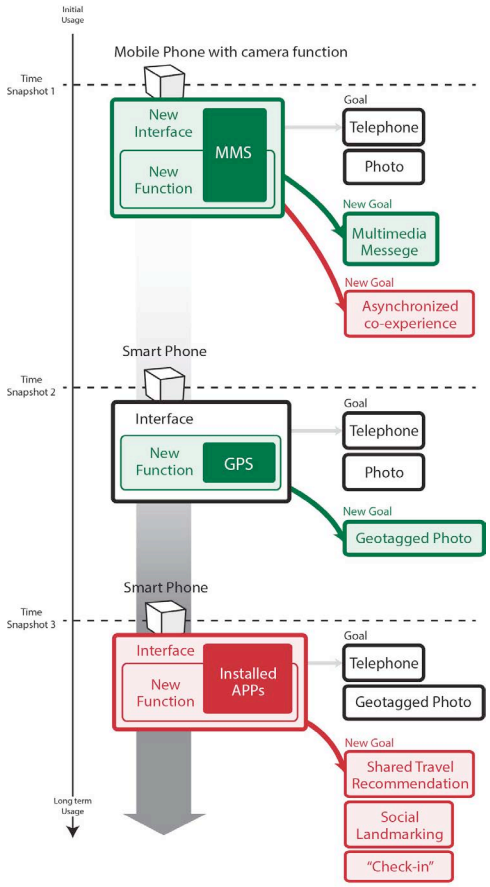


Figure 9: time snapshots using concept gestalt thinking

For example, after the initial introduction of mobile phones, a camera module was later embedded in the new generation of products. New mobile phone product lines added the function of photo taking as well as a new interface for users to take photos (ex: go the photo menu and choose "taking a photo"). It is a design feature to achieve the mobile photo taking and displayed in green color. At the same time, Multimedia Messaging Service (MMS) was also available, and photos can be sent out, just like ordinary text messages, to other mobile phones *(time snapshot 1)*.

Designers changed the goals of taking a photo with mobile phone to sharing photos with mobile phone to other users. However, users create their own innovative uses. MMS service became a platform of asynchronized co-experience [3]. Users may, for instance, send out a photo of a family gathering to family members who are not able to attend the gathering. Employees send out their current work to co-workers, such as sketches of ideas or craft in progress. Here users have created new uses for the photo functionality in mobile phones, which may be different from the designed goals (marked in red color).

When the mobile phone evolved into the smart phone, combined with several different internal functions and new interface, GPS became a new function directed related to mobile phone photo taking *(time snapshot 2)*. Since the smartphone keeps track of current users' geographic location, location data can be embedded into photos taken. One noticeable aspect is that the GPS function does not change the approach of mobile photography; it is an add-on function. With this new geotagged function, designers define use goals toward easy tracing of the place of taking photos.

Smartphone users then re-appropriate the geotagged photos toward a different use, rather than just a different way to sorting or showing photos *(time snapshot 3)*. They upload the photos, through installed APPs like Facebook, to recommend a place to visit, such as a favorite restaurant. The geotagged photos then provide a different goal that is more social in nature.

The reconstruction of mobile phone photography in Figure 9 demonstrates the evolution with three snapshots in time. The visual representation, arranged in a clear visual diagram, is a straightforward and intuitively direct way to convey the change in function, interface and goal. Through the color-coding, a reader can see, at various moment of usage, the intentions of both designers and users. In short, this visual schema *represents* the change in the past clearly, which can aid in better *communication* between design researchers and practitioners.

Discussion

We introduce examples where researchers carefully define concepts, particularly in a visual form, and use these concepts to analyze previous use experience. This can be seen as a bottom-up process, which has many similarities with text coding scheme in protocol analysis. We find the different ways that these visual approach concepts can help researchers to analyze users' sketches to be of the most interest.

We also present service blueprints and our historical review in a visual fashion. Through adding extra layer of information, concepts of interactive history, and snapshots of time, researchers can better *showcase* the trajectory of change.

All the examples we included here have one shared characteristic: they all *transform* the research and design thinking into being visually informed.

MERITS OF VISUAL APPRAOCHES IN LONGITUDINAL USERS RESEARCHES

In this section, we will in some detail discuss the potential merits of applying visual approaches, while in the next discuss some drawbacks. Here we will focus on two aspects. The first is that these approaches may support participation and communication. Secondly, visual approaches can help make sense of fragmented use data and information derived from longitudinal studies.

Encourage User's participation and communication in longitudinal user research

We argue that visual approaches can be intuitive and easy to use since they help to frame prior use scenarios and provide an alternative way to explain prior experience. The visualization of the time of use also becomes an anchored reference for discussion and communication within user research.

Chronological order and temporal framing

The coordination and diagrams applied in the examples we presented above helps users to situate themselves properly in relation to their prior use. It can be visually clear and understandable for the participants in user research to arrange different events from the left to the right, or from the top to the bottom, according to the visual form researchers provide. When research participants keep adding new events or annotating existing ones, prior inputs serve as reference points. The data can then help users reconsider the different relationships among these events of use already on the map or diagram. We believe that this naturally enables users to think about their experiences with a more holistic temporal perspective.

Alternative description of prior use experience

As we mentioned above, in various retrospective studies text is the main format for users to describe their prior use. However, drawing a line or making an entry on the diagram is simple and straightforward for many users. It breaks the language barriers mentioned above, that is, that people describe their prior experience with different depth, breadth, and terminology. Prior use experience that might be difficult to describe thoroughly can now be described as the increasing or decreasing of a line, or a turning point of the sketch to indicate certain events. The level of experience can, due to its visual quality, also be roughly compared among different sketches from users.

Timeline of use as platform of communication

We find that the "visual way" of retrospection and description is easy to understand, even for a general audience involved in the team research activities. For example, in our email experience study users were encouraged to explain their story of use to the researchers through their own sketches. We found that this was an intuitive way to invite insights from users regarding their prior use. When researchers need clarification from a research participant, it is also effective to simply refer to visual data and locate an event of prior use.

We have also introduced the historical analysis of mobile photography to research colleagues in conference and through public presentations. An immediate and common reaction from them is that they are eager to modify or enrich the current snapshots of events, by attaching related information to the current timeline. Although we only introduce and present the interaction through a fairly simple artifact centered perspective (function, interface and goal), the comments we receive are about adding critical events beyond the time snapshots we currently selected. Some common comments had to do with business aspects related to the events, threats from competitors, or comparison of two similar products.

We see this reaction as a sign that the visual and conceptual analysis can, most of all, serve as a *communication platform*. The design team or other stakeholders in the company can use it to understand the design process and a design strategy. Designers can now describe the use in the past, and showcase potential opportunities in the future. In sum, the visual description provides *temporal anchor points* that can further support detailed analysis and examinations.

Researchers Make Sense of Fragmented Prior Use Information

Collecting data from the users' retrospection in longitudinal studies inevitably produces fragmented pieces of use data. It is not always easy to develop an overall understanding from such data. For example, we might need to jump back and forth within a text description from different users, seek the relevant description and compare those similarities or differences. It also increases the difficulty for researchers to systematically analyze less coherent use data.

Can we *utilize heterogeneous and fragmented data of use* in a better way? Why do visual approaches become an effective alternatives procedure in longitudinal research? We believe that the notion of "*gestalt*" can be a relevant concept since it has been developed to understand wholeness from partials. The understanding of gestalt theory might help us to hypothesize why visual approaches have a positive influence over data analysis in long-term user researches.

The German, the word "*Gestalt*" indicates a form or shape, and in English it refers to a concept of wholeness. Gestalt principles, such as principles described in [27], are widely recognized in the visual design field and also applied in many interaction design practices, such as in [23]. Gestalt theory has developed a set of *principles*. For example, the more objects look similar to each other, the more they are perceived as a group (the principle of similarity). Continuation occurs when the eye moves through one object and continues to another object as a tendency (the principle of continuation). If a shape is not completely drawn and enough contours are revealed, humans tend to close the shape (the principle of closure). When shapes are placed close together, there are perceived as a group (the principle of proximity). The core concept of the gestalt rule is to see the *emergence of a whole from the parts*.

We find that the thinking activities, such as annotating prior events on the diagrams or linking those events with a line, are very similar to forming a "Gestalt" from parts. In other words, when we can only acquire partial and fragmented use information in the retrospective users research, the visual gestalt reasoning, either from users' side or from

researchers' side, helps to achieve a holistic understanding of prior use. We therefore believe that it would be a worthwhile undertaking to further develop not only the visual methods and their notations, but also principles and ideas on how to support gestalt thinking in relation to these methods.

DRAWBACKS OF VISUAL APPRAOCHES IN LONGITUDINAL USERS RESEARCHES

Not all research procedures are perfect: there are also drawbacks of introducing visual approaches in user studies. Here, we discuss two insights: first, visual approaches may lead to a too simplistic analysis; second, developing suitable concepts for visual representation is not always easy.

Over simplification of longitudinal use experience

In order to keep the visual representation simple and easy to use, the visual approaches we mentioned above have chosen specified measurements for prior experience, such as easy or difficult to use, relationship to the interactive artifacts, attention level, etc. When researchers decide to choose one aspect from the overall experience and present it in the visual approach, it has both positive and negative impacts: a single measurement helps the researcher to keep a clear focus, however, it at the same time limits the potential richness of the users expression regarding their use experience.

One possible work-around for this dilemma is to develop and use multiple visual representations measuring different aspects of user experiences. With the cross comparison of different visual representations, an overall tendency of use might emerge. Also, the text annotation on top of a sketch or extra space for written reflections might be also helpful for participants to reflect on their prior use with better breadth and depth.

Difficulty in forming visual concept for longitudinal use

It seems simple and intuitive for research participants to follow the visual approaches we mentioned above. However, the concepts behind those visual diagrams are somewhat difficult to develop while making it easily understandable by the research participants. For example, in our email study we choose attention level as the main concept in relation to the user's experience. Some research participants questioned this choice immediately and needed clarification of what "attention level" means in the email use experience.

Regarding the difficulties of developing concepts of use when it comes to the visual approaches, we plan to continue our research for various interactive products and system and see if general procedures or common sense or concepts related to temporality can be synthesized.

IMPLICATIONS: A COMPLEMENTARY RESEARCH PERSPECTIVE

Based on our research we do not argue that current user study approaches should be replaced by visual approaches. In fact each research method and procedure has its unique strength in relation to particular research goals. Instead, what we do suggest is that a complementary research perspective could be added to those already prevalent, such as experiments, observations, interviews, and other longitudinal user studies. By recognizing that retrospective information is always fragmented but also rich, we can apply visual thinking and thereby develop a better understanding of discontinuous and fragmented data. By exploring visual trends, we believe that approaches based on *gestalt* understanding of prior use experiences can become equally significant as research outcome from existing approaches.

CONCLUSION

In this paper, we review several innovative visual approaches aimed at supporting longitudinal user research aimed at capturing prolonged user experiences. We see our research as leading to two major contributions: first, our work raises the overall *awareness toward temporal issues* in HCI and interaction design research. We are confident that our proposed research agenda toward critical temporal issues is worthy to pursuit further. Second, we have reviewed the *merits and drawbacks of visual approaches* in longitudinal user studies. As a result we have concluded that visual approaches can serve as complementary perspectives when it comes to longitudinal user studies.

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